1. Questions

Study the following information carefully and answer the questions given below.

The table given below shows the total number of pens sold by two sellers, Seller A and Seller B, and the difference between the number of pens sold by Seller A and Seller B across five consecutive months in the year 2025.

Note: The number of pens sold by Seller A is greater than the number of pens sold by Seller B in all the given months.

Month	The total number of pens sold by Seller A and Seller B	The difference between the number of pens sold by Seller A and Seller B
January	800	200
February	1200	400
March	1600	400
April	1000	400
May	1800	600

If Seller B sold each pen for Rs. 50 in February, then find the total revenue generated by him during that month.

a. Rs. 15000

b. Rs. 18000

c. Rs. 20000

d. Rs. 24000

e. Rs. 25000

2. Questions

Find the ratio of the total number of pens sold by Seller A and Seller B together in March to the total number of pens sold by Seller A in February and April together.

a. 15: 14

b. 16: 15

c. 14: 13

d. 17: 16

e. 15: 19

3. Questions

If the ratio of red pens to black pens sold by Seller A in May was 7: 5, then find the number of black pens sold by him in that month.

Note: Consider only red and black pens sold by Seller A in May.

a. 500

b. 840

c. 600

d. 550

e. 700

4. Questions

The number of pens sold by Seller A in January is what percentage of the number of pens sold by Seller B in February?

a. 120%

b. 135%

c. 115%

d. 125%

e. 110%

5. Questions

Find the average number of pens sold by Seller B in January, March, and April.

a. 400

b. 500

c. 350

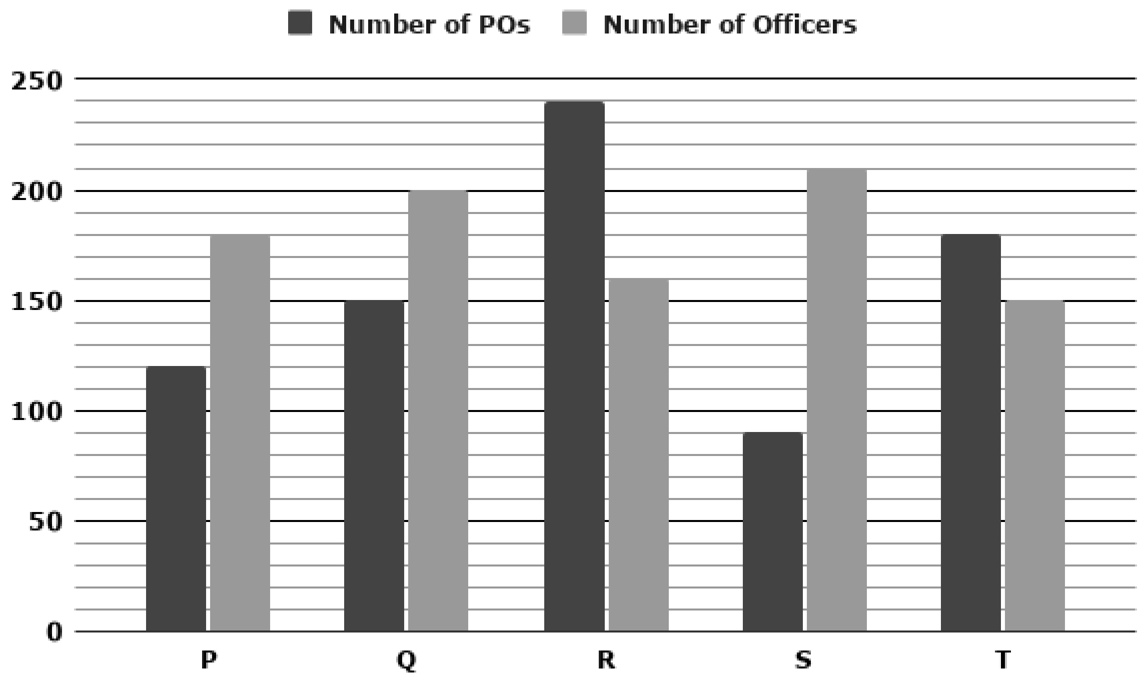
d. 450

e. 300

6. Questions

Study the following information carefully and answer the questions given below.

The bar graph given below represents the number of Probationary Officers (POs) and the number of Officers working in five different companies, namely P, Q, R, S, and T.



What is the ratio of the number of POs in companies P and R together to the number of Officers in companies P and S together?

a. 12 : 13

b. 11 : 13

c. 13 : 9

d. 12 : 17

e. 23 : 25

7. Questions

If the number of POs in company U is 40% more than that in company Q and the number of Officers in company U is 25% more than that in company Q, then find the total number of POs and Officers in company U.

a. 420

b. 440

c. 460

d. 480

e. 500

8. Questions

What is the difference between the total number of Officers in companies Q and R together and the total number of POs in companies Q and S together.

a. 110
b. 120
c. 130
d. 140
e. 150

9. Questions

What is the average number of Officers in companies Q, R, and T together?

a. 160
b. 165
c. 180
d. 175
e. 170

10. Questions

If the number of Clerks in company P is 75% more than the difference between the number of POs and Officers in company R, then find the number of Clerks in company P.

a. 120
b. 130
c. 140
d. 150
e. 160

11. Questions

A boat can travel 150 km upstream in 3 hours. If the speed of the stream is 1/5th of the speed of the boat upstream, how much time will the boat take to travel 350 km downstream?

a. 4 hours
b. 7 hours
c. 5 hours
d. 6 hours
e. 8 hours

12. Questions

The simple interest earned on Rs. 4000 at 10% per annum for 2 years is reinvested at a 20% per

annum compound interest, compounded half-yearly for one year. Find the total amount received on the reinvested sum.

a. Rs. 928

b. Rs. 968

c. Rs. 948

d. Rs. 988

e. Rs. 918

13. Questions

Pipe A can fill a tank in 4 hours and Pipe B can fill 1/3rd of the same tank in 2 hours. If both pipes are opened together, how much time will they take to fill the tank completely?

a. 2 hours 30 minutes

b. 3 hours 24 minutes

c. 1 hour 48 minutes

d. 2 hours 24 minutes

e. 3 hours 36 minutes

14. Questions

The length of a rectangle is 4 cm more than its breadth and the area of the rectangle is 96 cm^2. Find the perimeter of a square whose side is equal to the length of the rectangle.

a. 48 cm

b. 36 cm

c. 44 cm

d. 52 cm

e. 40 cm

15. Questions

A container contains 60 litres of a mixture of milk and water in the ratio 2 : 3, respectively. Some quantity of water is added to the mixture so that the ratio of milk to water becomes 1 : 2. Find the quantity of water added to the mixture.

a. 15 liters

b. 10 liters

c. 18 liters

d. 8 liters

e. 12 liters

16. Questions

A and B invested in a retail store with capital in the ratio of 4:5, respectively. Their investment periods were 9 months and 12 months, respectively. If the profit share of B is Rs. 1500, then find the profit share of A.

a. Rs. 1000

b. Rs. 1200

c. Rs. 800

d. Rs. 900

e. Rs. 1100

17. Questions

Sharvin's average marks in 5 examinations is 60. If his average marks in the first 3 examinations is 50 and that in the last 3 examinations is 75, then find the marks obtained by Sharvin in the third examination.

a. 65

b. 75

c. 80

d. 70

e. 85

18. Questions

Ragu purchased a cycle. He spent 10% of the purchase price on transportation and an additional 15% of the purchase price on repairs. If the total expenditure on transportation and repairs is Rs. 1,000, find the selling price of the cycle required to earn a profit of 20%.

a. Rs. 6000

b. Rs. 5500

c. Rs. 6400

d. Rs. 5800

e. Rs. 6200

19. Questions

A piggy bank contains Rs. 210 in the form of 1 rupee, 50 paise, and 75 paise coins in the ratio 1:2:2, respectively. Find the total number of coins in the piggy bank.

a. 280

b. 350

c. 300

d. 400

e. 250

20. Questions

The present age of P is 2/5 times the present age of Q. Three years ago, the age of P was 1/3 times the age of Q at that time. If R is 5 years older than Q, then find the present age of R.

a. 40 years

b. 30 years

c. 45 years

d. 25 years

e. 35 years

21. Questions

What value will come in place of the question mark (?) in the following question?

(3 / 8) * 640 + (5 / 12) * 1440 = ? + 150

a. 650

b. 670

c. 690

d. 710

e. 730

22. Questions

[(√1296 + √2025)/9] * 15 = ?

a. 115

b. 125

c. 145

d. 155

e. 135

23. Questions

24 * 16 + 32 * 14 = ? + 200

a. 622

b. 632

c. 642

d. 652

e. 612

24. Questions

$343^2 * 49^3 / 7^4 = 7^?$

a. 6

b. 8

c. 10

d. 12

e. 7

25. Questions

$(5/8) * 512 + 12^2 \div 8 = ? + 138$

a. 180

b. 210

c. 200

d. 190

e. 220

26. Questions

What approximate value will come in place of the question mark (?) in the following question?

39.95% of 599.98 + ?% of 249.95 = 339.92

a. 80

b. 10

c. 40

d. 50

e. 30

27. Questions

√1023.95 + √400.05 - 31.91 = ? * 3.99

a. 25

b. 5
c. 15
d. 50
e. 80

28. Questions

$$\frac{15.01^4}{4.98^4} + 2.99^2 = \sqrt{?}$$

a. 3600
b. 8100
c. 2500
d. 12100
e. 10000

29. Questions

$?^2 + 80.05 = 119.92 * 4.01$

a. 10
b. 40
c. 20
d. 30
e. 80

30. Questions

$(15.04 - \sqrt{35.95})^2 + (11.96 - \sqrt{63.92})^2 = ?$

a. 95
b. 102
c. 91
d. 85
e. 97

31. Questions

What value will come in the place of (?) in the following number series?

28, 77, ?, 279, 448, 673

a. 154
b. 158
c. 162
d. 168
e. 150

32. Questions

?, 402, 387, 369, 348, 324

a. 412
b. 416
c. 414
d. 418
e. 420

33. Questions

7, 13, 23, 41, 75, ?

a. 141
b. 145
c. 135
d. 139
e. 143

34. Questions

6, 12, 20, ?, 42, 56

a. 30
b. 32
c. 28
d. 34
e. 36

35. Questions

15, ?, 20, 25, 32, 43

a. 16

b. 18

c. 19

d. 17

e. 21

36. Questions

Find the wrong number in the following series.

121, 134, 153, 170, 193, 222

a. 134

b. 170

c. 153

d. 193

e. 222

37. Questions

571, 553, 533, 511, 489, 461

a. 489

b. 553

c. 533

d. 511

e. 461

38. Questions

2880, 490, 96, 24, 8, 4

a. 96

b. 24

c. 4

d. 490

e. 8

39. Questions

45, 49, 58, 76, 99, 135

a. 58

b. 99

c. 76

d. 49

e. 135

40. Questions

15, 20, 27, 37, 51, 72

a. 51

b. 20

c. 27

d. 72

e. 37

41. Questions

In each of the following questions, two equations are given. You have to solve both equations to find the relation between x and y.

I). $x^2 + 12x - 64 = 0$

II). $y^2 - 18y + 56 = 0$

a. $x > y$

b. $x \geq y$

c. $x < y$

d. $x \leq y$

e. x = y or relationship can't be determined.

42. Questions

I). $12x^2 = 108$

II). $y^2 + 11y + 30 = 0$

a. $x > y$

b. $x \geq y$

c. $x < y$

d. $x \leq y$

e. x = y or the relationship can't be determined.

43. Questions

I). $x^2 + 23x + 132 = 0$

II). $y^2 + 4y - 60 = 0$

a. $x > y$

b. $x \geq y$

c. $x < y$

d. $x \leq y$

e. $x = y$ or the relationship can't be determined.

44. Questions

I). $3x + 4y = 35$

II). $2x - y = 5$

a. $x > y$

b. $x \geq y$

c. $x < y$

d. $x \leq y$

e. $x = y$ or the relationship can't be determined.

45. Questions

I). $x^2 - 14x + 48 = 0$

II). $y^2 - 9y + 18 = 0$

a. $x > y$

b. $x \geq y$

c. $x < y$

d. $x \leq y$

e. $x = y$ or the relationship can't be determined.

Explanations:

1. Questions

We can apply the definitive formulas:

Seller A = (Total + Difference) / 2

Seller B = (Total - Difference) / 2

In January:

The total number of pens sold by Seller A and Seller B in January = 800

The difference between the number of pens sold by Seller A and Seller B in January = 200

The number of pens sold by Seller A in January = (800 + 200) / 2 = 500

The number of pens sold by Seller B in January = (800 - 200) / 2 = 300

In February:

The total number of pens sold by Seller A and Seller B in February = 1200

The difference between the number of pens sold by Seller A and Seller B in February = 400

The number of pens sold by Seller A in February = (1200 + 400) / 2 = 800

The number of pens sold by Seller B in February = (1200 - 400) / 2 = 400

In March:

The total number of pens sold by Seller A and Seller B in March = 1600

The difference between the number of pens sold by Seller A and Seller B in March = 400

The number of pens sold by Seller A in March = (1600 + 400) / 2 = 1000

The number of pens sold by Seller B in March = (1600 - 400) / 2 = 600

In April:

The total number of pens sold by Seller A and Seller B in April = 1000

The difference between the number of pens sold by Seller A and Seller B in April = 400

The number of pens sold by Seller A in April = (1000 + 400) / 2 = 700

The number of pens sold by Seller B in April = (1000 - 400) / 2 = 300

In May:

The total number of pens sold by Seller A and Seller B in May = 1800

The difference between the number of pens sold by Seller A and Seller B in May = 600

The number of pens sold by Seller A in May = (1800 + 600) / 2 = 1200

The number of pens sold by Seller B in May = (1800 - 600) / 2 = 600

Final Value Table:

Month	The number of pens sold by Seller A	The number of pens sold by Seller B
January	500	300
February	800	400
March	1000	600
April	700	300
May	1200	600

Answer: C

From the final value table,

The number of pens sold by Seller B in February = 400

Selling price of each pen = Rs. 50

Total revenue generated by Seller B in February = 400 * 50 = Rs. 20000

Hence, the correct answer is option C.

2. Questions

We can apply the definitive formulas:

Seller A = (Total + Difference) / 2

Seller B = (Total - Difference) / 2

In January:

The total number of pens sold by Seller A and Seller B in January = 800

The difference between the number of pens sold by Seller A and Seller B in January = 200

The number of pens sold by Seller A in January = (800 + 200) / 2 = 500

The number of pens sold by Seller B in January = (800 - 200) / 2 = 300

In February:

The total number of pens sold by Seller A and Seller B in February = 1200

The difference between the number of pens sold by Seller A and Seller B in February = 400

The number of pens sold by Seller A in February = (1200 + 400) / 2 = 800

The number of pens sold by Seller B in February = (1200 - 400) / 2 = 400

In March:

The total number of pens sold by Seller A and Seller B in March = 1600

The difference between the number of pens sold by Seller A and Seller B in March = 400

The number of pens sold by Seller A in March = (1600 + 400) / 2 = 1000

The number of pens sold by Seller B in March = (1600 - 400) / 2 = 600

In April:

The total number of pens sold by Seller A and Seller B in April = 1000

The difference between the number of pens sold by Seller A and Seller B in April = 400

The number of pens sold by Seller A in April = (1000 + 400) / 2 = 700

The number of pens sold by Seller B in April = (1000 - 400) / 2 = 300

In May:

The total number of pens sold by Seller A and Seller B in May = 1800

The difference between the number of pens sold by Seller A and Seller B in May = 600

The number of pens sold by Seller A in May = (1800 + 600) / 2 = 1200

The number of pens sold by Seller B in May = (1800 - 600) / 2 = 600

Final Value Table:

Month	The number of pens sold by Seller A	The number of pens sold by Seller B
January	500	300
February	800	400
March	1000	600
April	700	300
May	1200	600

Answer: B

From the final value table,

The total number of pens sold by Seller A and Seller B together in March = 1000 + 600 = 1600

The total number of pens sold by Seller A in February and April together = 800 + 700 = 1500

Required ratio = 1600 : 1500

Required ratio = 16 : 15

Hence, the correct answer is option B.

3. Questions

We can apply the definitive formulas:

Seller A = (Total + Difference) / 2

Seller B = (Total - Difference) / 2

In January:

The total number of pens sold by Seller A and Seller B in January = 800

The difference between the number of pens sold by Seller A and Seller B in January = 200

The number of pens sold by Seller A in January = (800 + 200) / 2 = 500

The number of pens sold by Seller B in January = (800 - 200) / 2 = 300

In February:

The total number of pens sold by Seller A and Seller B in February = 1200

The difference between the number of pens sold by Seller A and Seller B in February = 400

The number of pens sold by Seller A in February = (1200 + 400) / 2 = 800

The number of pens sold by Seller B in February = (1200 - 400) / 2 = 400

In March:

The total number of pens sold by Seller A and Seller B in March = 1600

The difference between the number of pens sold by Seller A and Seller B in March = 400

The number of pens sold by Seller A in March = (1600 + 400) / 2 = 1000

The number of pens sold by Seller B in March = (1600 - 400) / 2 = 600

In April:

The total number of pens sold by Seller A and Seller B in April = 1000

The difference between the number of pens sold by Seller A and Seller B in April = 400

The number of pens sold by Seller A in April = (1000 + 400) / 2 = 700

The number of pens sold by Seller B in April = (1000 - 400) / 2 = 300

In May:

The total number of pens sold by Seller A and Seller B in May = 1800

The difference between the number of pens sold by Seller A and Seller B in May = 600

The number of pens sold by Seller A in May = (1800 + 600) / 2 = 1200

The number of pens sold by Seller B in May = (1800 - 600) / 2 = 600

Final Value Table:

Month	The number of pens sold by Seller A	The number of pens sold by Seller B
January	500	300
February	800	400
March	1000	600
April	700	300
May	1200	600

Answer: A

From the final value table,

The total number of pens sold by Seller A in May = 1200

The ratio of red pens to black pens sold = 7 : 5

Total parts = 7 + 5 = 12

The number of black pens sold = 1200 * (5 / 12) = 100 * 5 = 500

Hence, the correct answer is option A.

4. Questions

We can apply the definitive formulas:

Seller A = (Total + Difference) / 2

Seller B = (Total - Difference) / 2

In January:

The total number of pens sold by Seller A and Seller B in January = 800

The difference between the number of pens sold by Seller A and Seller B in January = 200

The number of pens sold by Seller A in January = (800 + 200) / 2 = 500

The number of pens sold by Seller B in January = (800 - 200) / 2 = 300

In February:

The total number of pens sold by Seller A and Seller B in February = 1200

The difference between the number of pens sold by Seller A and Seller B in February = 400

The number of pens sold by Seller A in February = (1200 + 400) / 2 = 800

The number of pens sold by Seller B in February = (1200 - 400) / 2 = 400

In March:

The total number of pens sold by Seller A and Seller B in March = 1600

The difference between the number of pens sold by Seller A and Seller B in March = 400

The number of pens sold by Seller A in March = (1600 + 400) / 2 = 1000

The number of pens sold by Seller B in March = (1600 - 400) / 2 = 600

In April:

The total number of pens sold by Seller A and Seller B in April = 1000

The difference between the number of pens sold by Seller A and Seller B in April = 400

The number of pens sold by Seller A in April = (1000 + 400) / 2 = 700

The number of pens sold by Seller B in April = (1000 - 400) / 2 = 300

In May:

The total number of pens sold by Seller A and Seller B in May = 1800

The difference between the number of pens sold by Seller A and Seller B in May = 600

The number of pens sold by Seller A in May = (1800 + 600) / 2 = 1200

The number of pens sold by Seller B in May = (1800 - 600) / 2 = 600

Final Value Table:

Month	The number of pens sold by Seller A	The number of pens sold by Seller B
January	500	300
February	800	400
March	1000	600
April	700	300
May	1200	600

Answer: D

From the final value table,

The number of pens sold by Seller A in January = 500

The number of pens sold by Seller B in February = 400

Required percentage = (500 / 400) * 100

Required percentage = (5 / 4) * 100 = 125%

Hence, the correct answer is option D.

5. Questions

We can apply the definitive formulas:

Seller A = (Total + Difference) / 2

Seller B = (Total - Difference) / 2

In January:

The total number of pens sold by Seller A and Seller B in January = 800

The difference between the number of pens sold by Seller A and Seller B in January = 200

The number of pens sold by Seller A in January = (800 + 200) / 2 = 500

The number of pens sold by Seller B in January = (800 - 200) / 2 = 300

In February:

The total number of pens sold by Seller A and Seller B in February = 1200

The difference between the number of pens sold by Seller A and Seller B in February = 400

The number of pens sold by Seller A in February = (1200 + 400) / 2 = 800

The number of pens sold by Seller B in February = (1200 - 400) / 2 = 400

In March:

The total number of pens sold by Seller A and Seller B in March = 1600

The difference between the number of pens sold by Seller A and Seller B in March = 400

The number of pens sold by Seller A in March = (1600 + 400) / 2 = 1000

The number of pens sold by Seller B in March = (1600 - 400) / 2 = 600

In April:

The total number of pens sold by Seller A and Seller B in April = 1000

The difference between the number of pens sold by Seller A and Seller B in April = 400

The number of pens sold by Seller A in April = (1000 + 400) / 2 = 700

The number of pens sold by Seller B in April = (1000 - 400) / 2 = 300

In May:

The total number of pens sold by Seller A and Seller B in May = 1800

The difference between the number of pens sold by Seller A and Seller B in May = 600

The number of pens sold by Seller A in May = (1800 + 600) / 2 = 1200

The number of pens sold by Seller B in May = (1800 - 600) / 2 = 600

Final Value Table:

Month	The number of pens sold by Seller A	The number of pens sold by Seller B
January	500	300
February	800	400
March	1000	600
April	700	300
May	1200	600

Answer: A

The number of pens sold by Seller B in January = 300

The number of pens sold by Seller B in March = 600

The number of pens sold by Seller B in April = 300

Total number of pens sold in these three months = 300 + 600 + 300 = 1200

Required Average = 1200 / 3 = 400

Hence, the correct answer is option A.

6. Questions

Final value table,

Company	Number of POs	Number of Officers
P	120	180
Q	150	200
R	240	160
S	90	210
T	180	150

Answer: A

From the final value table,

The total number of POs in companies P and R together = 120 + 240 = 360

The total number of Officers in companies P and S together = 180 + 210 = 390

Required ratio = 360 : 390

Required ratio = 12 : 13

Hence, the correct answer is option A.

7. Questions

Final value table,

Company	Number of POs	Number of Officers
P	120	180
Q	150	200
R	240	160
S	90	210
T	180	150

Answer: C

From the final value table,

The number of POs in company Q = 150

The number of POs in company U = 150 + 150 * (40 / 100) = 150 + 60 = 210

The number of Officers in company Q = 200

The number of Officers in company U = 200 + 200 * (25 / 100) = 200 + 50 = 250

Total number of POs and Officers in company U = 210 + 250 = 460

Hence, the correct answer is option C.

8. Questions

Final value table,

Company	Number of POs	Number of Officers
P	120	180
Q	150	200
R	240	160
S	90	210
T	180	150

Answer: B

From the final value table,

The total number of Officers in companies Q and R together = 200 + 160 = 360

The total number of POs in companies Q and S together = 150 + 90 = 240

Required difference = 360 - 240 = 120

Hence, the correct answer is option B.

9. Questions

Final value table,

Company	Number of POs	Number of Officers
P	120	180
Q	150	200
R	240	160
S	90	210
T	180	150

Answer: E

From the final value table,

The total number of Officers in companies Q, R, and T = (200 + 160 + 150) = 510

Required Average = 510 / 3 = 170

Hence, the correct answer is option E.

10. Questions

Final value table,

Company	Number of POs	Number of Officers
P	120	180
Q	150	200
R	240	160
S	90	210
T	180	150

Answer: C

From the final value table,

The number of POs in company R = 240

The number of Officers in company R = 160

Difference between the number of POs and Officers in company R = 240 - 160 = 80

The number of Clerks in company P = 175% of 80 = (175 / 100) * 80

The number of Clerks in company P = (7 / 4) * 80 = 140

Hence, the correct answer is option C.

11. Questions

Answer: C

Let the speed of the boat in still water be 'u' and the speed of the stream be 'v'.

Upstream speed = u - v

u - v = 150 / 3 = 50 km/hr

Speed of the stream,

v = 50 * (1 / 5) = 10 km/hr

Speed of the boat in still water,

u - 10 = 50

u = 60 km/hr

Downstream speed = u + v

u + v = 60 + 10 = 70 km/hr

Time taken to travel downstream = 350 / 70 = 5 hours

Hence, the correct answer is option C.

12. Questions

Answer: B

Let Simple Interest be SI.

SI = $P_1 * N_1 * (R_1 / 100)$

SI = (4000 * 2 * 10) / 100 = Rs. 800

Principal for compound interest (P_2) = Rs. 800

Rate of interest half-yearly (R_2) = 20% / 2 = 10%

Number of half-years in 1 year (n) = 1 * 2 = 2

Amount = $P_2 * [1 + (R_2/100)]^n$

Amount = $800 * [1 + (10/100)]^2$

⇒ 800 * (110 / 100) * (110 / 100)

⇒ 8 * 11 * 11

⇒ 968

Amount = Rs. 968

Hence, the correct answer is option B.

13. Questions

Answer: D

Time taken by A to fill the entire tank = 4 hours

Time taken by B to fill the entire tank = 2 * 3 = 6 hours

Let the total capacity of the tank = LCM of (4, 6) = 12 units

Efficiency of A = 12 / 4 = 3 units/hour

Efficiency of B = 12 / 6 = 2 units/hour

Total efficiency = 3 + 2 = 5 units/hour

Time taken = Total capacity / Total efficiency

Time taken = 12 / 5 = 2.4 hours (or) 2 hours 24 minutes

Hence, the correct answer is option D.

14. Questions

Answer: A

Let the breadth of the rectangle be ‘b’ cm.

Length of the rectangle = (b + 4) cm

Area = Length * Breadth = 96 cm^2

According to the problem,

(b + 4) * b = 96

$b^2 + 4b - 96 = 0$

$b^2 + 12b - 8b - 96 = 0$

b (b + 12) - 8 (b + 12) = 0

(b + 12) (b - 8) = 0

b = +8 or -12

b = 8 cm (Since the breadth is always positive)

Length = 8 + 4 = 12 cm

Side of the square (a) = Length = 12 cm

Perimeter of the square = 4 * a

Perimeter of the square = 4 * 12 = 48 cm

Hence, the correct answer is option A.

15. Questions

Answer: E

Let the initial quantities of **milk** and **water** be **M** litres and **W** litres, respectively.

M = 60 * (2 / 5) = 24 liters

W = 60 * (3 / 5) = 36 liters

Let the quantity of water added be ‘x’ litres.

Then,

[24 / (36 + x)] = 1 / 2

48 = 36 + x

x = 48 - 36 = 12 liters

Hence, the correct answer is option E.

16. Questions

Answer: D

Let investment of A and B be Rs. 4x and Rs. 5x.

Time period of A and B = 9 months and 12 months.

Profit ratio of A and B = (4x * 9) : (5x * 12)

Profit ratio = 36x : 60x = 3 : 5

Let the profit shares of A and B be Rs. 3y and Rs. 5y, respectively.

Given that, Profit share of B = Rs. 1500

5y = 1500

y = 1500 / 5 = 300

Profit share of A = 3y = 3 * 300 = Rs. 900

Hence, the correct answer is option D.

17. Questions

Answer: B

Let the marks obtained by Sharvin in the five examinations be a, b, c, d, and e.

His total marks in 5 examinations = 5 * 60 = 300

So, a + b + c + d + e = 300 …(i)

His total marks in the first 3 examinations = 3 * 50 = 150

So, a + b + c = 150

His total marks in the last 3 examinations = 3 * 75 = 225

So, c + d + e = 225

Now adding his 1st three and last three examinations marks

(a + b + c) + (c + d + e) = 150 + 225 = 375

(a + b + c + d + e) + c = 375…..(ii)

Using the value of equation (i) in equation (ii)

300 + c = 375

c = 75

Marks secured in the third examination = 75

Hence, the correct answer is option B.

18. Questions

Answer: A

Let the purchase price of the cycle be Rs. P.

Additional Expenditure percentage = 10% + 15% = 25%

Given,

25% of P = 1000

P = 1000 * (100 / 25) = Rs. 4000

Total cost price = 4000 + 1000 = Rs. 5000

To earn a profit of 20%,

Selling price = 120% of 5000 = Rs. 6000

Hence, the correct answer is option A.

19. Questions

Answer: C

Let the number of 1 rupee, 50 paise, and 75 paise coins be x, 2x, and 2x.

Total value in rupees = (x * 1) + (2x * 0.5) + (2x * 0.75) = 210

⇒ x + x + 1.5x = 210

⇒ 3.5x = 210

⇒ x = 210 / 3.5

⇒ x = 60

Total number of coins in the piggy bank = x + 2x + 2x = 5x

Total number of coins in the piggy bank = 5 * 60 = 300

Hence, the correct answer is option C.

20. Questions

Answer: E

Let the present ages of P and Q be 2x and 5x years, respectively.

Three years ago:

[(2x - 3) / (5x - 3)] = 1 / 3

6x - 9 = 5x - 3

x = 6

Therefore,

The present age of Q = 5 * 6 = 30 years

The present age of R = 30 + 5 = 35 years

Hence, the correct answer is option E.

21. Questions

Answer: C

⇒ (3 / 8) * 640 + (5 / 12) * 1440 = ? + 150

⇒ (3 * 80) + (5 * 120) = ? + 150

⇒ 240 + 600 = ? + 150

⇒ 840 = ? + 150

⇒ ? = 840 - 150

⇒ ? = 690

Hence, the correct answer is option C.

22. Questions

Answer: E

⇒ [(√1296 + √2025)/9] * 15 = ?

⇒ [(36 + 45)/9] * 15 = ?

⇒ (81/9) * 15 = ?

⇒ 9 * 15 = ?

⇒ ? = 135

Hence, the correct answer is option E.

23. Questions

Answer: B

⇒ (24 * 16) + (32 * 14) = ? + 200

⇒ 384 + 448 = ? + 200

⇒ 832 = ? + 200

⇒ ? = 832 - 200

⇒ ? = 632

Hence, the correct answer is option B.

24. Questions

Answer: B

⇒ $343^2 * 49^3 / 7^4 = 7^?$

⇒ $(7^3)^2 * (7^2)^3 / 7^4 = 7^?$

⇒ $7^6 * 7^6 / 7^4 = 7^?$

⇒ $7^{(6+6-4)} = 7^?$

⇒ $7^8 = 7^?$

⇒ ? = 8

Hence, the correct answer is option B.

25. Questions

Answer: C

⇒ (5/8) * 512 + 12^2 ÷ 8 = ? + 138

⇒ (5 * 64) + 144 / 8 = ? + 138

⇒ 320 + 18 = ? + 138

⇒ 338 = ? + 138

⇒ ? = 338 - 138

⇒ ? = 200

Hence, the correct answer is option C.

26. Questions

Answer: C

⇒ 39.95% of 599.98 + ?% of 249.95 = 339.92

⇒ 40% of 600 + ?% of 250 = 340

⇒ (40 / 100) * 600 + (? / 100) * 250 = 340

⇒ 240 + 2.5 * ? = 340

⇒ 2.5 * ? = 340 - 240

⇒ 2.5 * ? = 100

⇒ ? = 100 / 2.5

⇒ ? = 40

Hence, the correct answer is option C.

27. Questions

Answer: B

⇒ √1023.95 + √400.05 - 31.91 = ? * 3.99

⇒ √1024 + √400 - 32 = ? * 4

⇒ 32 + 20 - 32 = ? * 4

⇒ 20 = ? * 4

⇒ ? = 20 / 4

⇒ ? = 5

Hence, the correct answer is option B.

28. Questions

Answer: B

⇒ $15.01^4 \div 4.98^4 + 2.99^2 = \sqrt{?}$

⇒ $15^4 \div 5^4 + 3^2 = \sqrt{?}$

⇒ $(15 / 5)^4 + 3^2 = \sqrt{?}$

⇒ $3^4 + 3^2 = \sqrt{?}$

⇒ $81 + 9 = \sqrt{?}$

⇒ $90 = \sqrt{?}$

⇒ $? = 90^2$

⇒ $? = 8100$

Hence, the correct answer is option B.

29. Questions

Answer: C

⇒ $?^2 + 80.05 = 119.92 * 4.01$

⇒ $?^2 + 80 = 120 * 4$

⇒ $?^2 + 80 = 480$

⇒ $?^2 = 480 - 80$

⇒ $?^2 = 400$

⇒ $? = 20$

Hence, the correct answer is option C.

30. Questions

Answer: E

⇒ $(15.04 - \sqrt{35.95})^2 + (11.96 - \sqrt{63.92})^2 = ?$

⇒ $(15 - \sqrt{36})^2 + (12 - \sqrt{64})^2 = ?$

⇒ $(15 - 6)^2 + (12 - 8)^2 = ?$

⇒ $9^2 + 4^2 = ?$

⇒ $81 + 16 = ?$

⇒ $? = 97$

Hence, the correct answer is option E.

31. Questions

Answer: B

The series follows the pattern of addition of squares of consecutive odd numbers starting from 7:

$28 + 7^2 = 77$

$77 + 9^2 =$ **158**

$158 + 11^2 = 279$

$279 + 13^2 = 448$

$448 + 15^2 = 673$

So, ? = 158.

Hence, the correct answer is option B.

32. Questions

Answer: C

The series follows the pattern of subtracting consecutive multiples of 3 starting from 12:

? - 12 = 402

So, ? = 414.

414 - 12 = 402

402 - 15 = 387

387 - 18 = 369

369 - 21 = 348

348 - 24 = 324

Hence, the correct answer is option C.

33. Questions

Answer: A

The series follows the pattern of (* 2 - 1), (* 2 - 3), (* 2 - 5), (* 2 - 7), (* 2 - 9):

7 * 2 - 1 = 13

13 * 2 - 3 = 23

23 * 2 - 5 = 41

41 * 2 - 7 = 75

75 * 2 - 9 = **141**

Or

7 13 23 41 75 141

6 10 18 34 66

4 8 16 32

So, ? = 141.

Hence, the correct answer is option A.

34. Questions

Answer: A

The series follows the pattern of (n * (n+1)) where n starts from 2:

2 * 3 = 6

3 * 4 = 12

4 * 5 = 20

5 * 6 = **30**

6 * 7 = 42

7 * 8 = 56

So, ? = 30.

OR

The series follows a pattern of adding consecutive even numbers starting from 6.

6 + 6 = 12

12 + 8 = 20

20 + 10 = **30**

30 + 12 = 42

42 + 14 = 56

Hence, the correct answer is option A.

35. Questions

Answer: D

The series follows the pattern of addition of consecutive prime numbers:

15 + 2 = **17**

17 + 3 = 20

20 + 5 = 25

25 + 7 = 32

32 + 11 = 43

So, ? = 17.

Hence, the correct answer is option D.

36. Questions

Answer: C

121 + 13 = 134

134 + 17 = **151** (Therefore, 153 is wrong and should be 151)

151 + 19 = 170

170 + 23 = 193

193 + 29 = 222

Logic: Difference is consecutive prime numbers starting from 13 (13, 17, 19, 23, 29).

Hence, the correct answer is option C.

37. Questions

Answer: A

571 - 18 = 553

553 - 20 = 533

533 - 22 = 511

511 - 24 = **487** (Therefore, 489 is wrong and should be 487)

487 - 26 = 461

Logic: Difference is consecutive even numbers being subtracted (18, 20, 22, 24, 26).

Hence, the correct answer is option A.

38. Questions

Answer: D

2880 ÷ 6 = **480** (Therefore, 490 is wrong and should be 480)

480 ÷ 5 = 96

96 ÷ 4 = 24

24 ÷ 3 = 8

8 ÷ 2 = 4

Logic: Terms are divided by consecutive decreasing numbers (6, 5, 4, 3, 2).

Hence, the correct answer is option D.

39. Questions

Answer: C

$45 + 2^2 = 49$

$49 + 3^2 = 58$

$58 + 4^2 =$ **74** (Therefore, 76 is wrong and should be 74)

$74 + 5^2 = 99$

$99 + 6^2 = 135$

Logic: Difference is the squares of consecutive natural numbers starting from 2.

Hence, the correct answer is option C.

40. Questions

Answer: D

15 20 27 37 51 **70**

5 7 10 14 19

2 3 4 5

(Therefore, 72 is wrong and should be 70)

Logic: The differences between consecutive terms are 5, 7, 10, 14, 19. The double differences are consecutive numbers starting from 2 (2, 3, 4, 5).

Hence, the correct answer is option D.

41. Questions

Answer: D

I). $x^2 + 12x - 64 = 0$

$x^2 + 16x - 4x - 64 = 0$

$x (x + 16) - 4 (x + 16) = 0$

$(x + 16) (x - 4) = 0$

$x = +4, -16$

II). $y^2 - 18y + 56 = 0$

$y^2 - 14y - 4y + 56 = 0$

$y (y - 14) - 4 (y - 14) = 0$

$(y - 14) (y - 4) = 0$

$y = +14, +4$

Comparison:

The value of x	The value of y	Comparison
+4	+14	x < y
+4	+4	x = y
-16	+14	x < y
-16	+4	x < y

Therefore, $x \leq y$.

Hence, the correct answer is option D.

42. Questions

Answer: A

I). $12x^2 = 108$

$x^2 = 9$

$x = +3, -3$

II). $y^2 + 11y + 30 = 0$

$y^2 + 5y + 6y + 30 = 0$

$y (y + 5) + 6 (y + 5) = 0$

$(y + 5) (y + 6) = 0$

$y = -5, -6$

Comparison:

The value of x	The value of y	Comparison
+3	-5	x > y
+3	-6	x > y
-3	-5	x > y
-3	-6	x > y

Therefore, $x > y$.

Hence, the correct answer is option A.

43. Questions

Answer: C

I). $x^2 + 23x + 132 = 0$

$x^2 + 11x + 12x + 132 = 0$

$x (x + 11) + 12 (x + 11) = 0$

$(x + 11) (x + 12) = 0$

$x = -11, -12$

II). $y^2 + 4y - 60 = 0$

$y^2 - 6y + 10y - 60 = 0$

$y (y - 6) + 10 (y - 6) = 0$

$(y - 6) (y + 10) = 0$

$y = +6, -10$

Comparison:

The value of x	The value of y	Comparison
-11	+6	x < y
-11	-10	x < y
-12	+6	x < y
-12	-10	x < y

Therefore, $x < y$.

Hence, the correct answer is option C.

44. Questions

Answer: E

I). $3x + 4y = 35$

II). $2x - y = 5$

Multiply equation II by 4:

$8x - 4y = 20$

Add equation I and new equation II:

$(3x + 4y) + (8x - 4y) = 35 + 20$

$11x = 55$

x = +5

Substitute x = 5 in equation II:

2(5) - y = 5

10 - y = 5

y = +5

Comparison:

+5 = +5

Therefore, x = y.

Hence, the correct answer is option E.

45. Questions

Answer: B

I). $x^2 - 14x + 48 = 0$

$x^2 - 8x - 6x + 48 = 0$

x (x - 8) - 6 (x - 8) = 0

(x - 8) (x - 6) = 0

x = +8, +6

II). $y^2 - 9y + 18 = 0$

$y^2 - 6y - 3y + 18 = 0$

y (y - 6) - 3 (y - 6) = 0

(y - 6) (y - 3) = 0

y = +6, +3

Comparison:

The value of x	The value of y	Comparison
+8	+6	x > y
+8	+3	x > y
+6	+6	x = y
+6	+3	x > y

Therefore, x ≥ y

Hence, the correct answer is option B.